

Why does L1 regularization give rise to sparse values of parameters.

L2 regularized obj

$$\min_w L(w) + \gamma \|w\|_2^2$$

$\Leftrightarrow$

$$\min_w L(w) \text{ s.t. } \|w\|_2 \leq \sqrt{\frac{\gamma}{\lambda}}$$

L1 regularizing objective

$$\min_w L(w) + \gamma \|w\|_1$$

$$\min_w L(w) \text{ s.t. } \|w\|_1 \leq \frac{\gamma}{\lambda}$$

observe pictorially

$$F(w) = \min_w \underbrace{\sum_{i=1}^N \frac{1}{2} (w^T x^i - y_i)^2}_{L(w)} + \lambda \|w\|_n$$

$$\frac{\partial L(w)}{\partial w_j}$$

$$= \sum_{i=1}^N (w^T x^i - y_i) x_j^i$$

$$= \sum_{i=1}^N (\underbrace{w_{-j}^T x_{-j}^i}_{\text{residual}} + w_j x_j^i - y_i) x_j^i$$

$$= \sum_{i=1}^N x_j^i (\underbrace{y_i - w_{-j}^T x_{-j}^i}_{\text{residual}}) + \sum_{i=1}^N w_j (x_j^i)^2$$

residual of "y" that is not covered by other dimension

$$w = \begin{bmatrix} w_1 \\ \vdots \\ w_d \end{bmatrix}$$

$w_{-j} \equiv w$ with just j th component removed

$$= \underbrace{\sum_{i=1}^N x_j^i \cdot \underbrace{x_{-j}^i}_{\text{residual}}}_{\text{correlation between } x_j \text{ \& residual}} + w_j \underbrace{\sum_{i=1}^N (x_j^i)^2}_{a_j}$$

$$\boxed{x - \hat{x}}$$

So the overall gradient of F

$$\frac{\partial F}{\partial w_j} = \frac{a_j w_j - c_j + \lambda \frac{\partial \|W\|_n}{\partial w_j}}{\partial w_j}$$

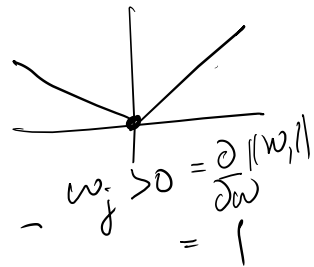
if $n=2$: L2 regularization:

$$\frac{\partial F}{\partial w_j} = a_j w_j - c_j + 2\lambda w_j = 0 \Rightarrow \underline{\underline{w_j^*}} = \frac{c_j}{a_j + 2\lambda}$$

if $n=1$ (L1 regularization)

$$\underline{\underline{\frac{\partial F}{\partial w_j}}} = \begin{cases} a_j w_j - c_j + \lambda & \text{if } w_j > 0 \\ a_j w_j - c_j - \lambda & \text{if } w_j < 0 \end{cases}$$

$$\|W_1\| = \sum_{j=1}^d |w_j|$$



$$\underline{\underline{[a_j w_j - c_j - \lambda, a_j w_j - c_j + \lambda]}} \text{ if } w_j = 0$$

w_j^* is global minima if 0 is one of the sub-gradients of F at w_j^*

$0 \in \frac{\partial F}{\partial w_j^*} \Rightarrow w_j = \begin{cases} \frac{c_j - \lambda}{a_j} & \text{if } c_j > \lambda \rightarrow \text{positive correlation} \\ \frac{c_j + \lambda}{a_j} & \text{if } c_j < -\lambda \rightarrow \text{negative correlation} \\ 0 & \text{if } -\lambda \leq c_j \leq \lambda \end{cases}$

$\underline{0}$

Probabilistic interpretation of linear regression

Motivation for probabilistic models

So far

- We assume a model outputs a predicted value. —
- During training we define a loss function that minimizes error of predicted value with respect to the true value in the training data
- Overfitting is controlled by minimizing error on unseen data

Limitation:

- Does not express uncertainty of predicted value
- No principled guideline for choice of loss function
- Overfitting control seems ad hoc.

Probabilistic models seek to remove this limitation

Probabilistic models

- Model outputs a distribution over possible values instead of a single value

$$x \rightarrow \boxed{f} \rightarrow p(y|x) \quad \hat{y} = \underset{y}{\operatorname{argmax}} p(y|x)$$

Example for linear regression

$$P(y|x) = \mathcal{N}(\underline{y}; \underbrace{\mu_y = f(x)}_{\text{learned}}; \underbrace{\sigma^2}_{\text{fixed}})$$

Linear regression

$$f(x) = w^T x \Rightarrow \text{we learn } w.$$

Training objective

Given $D = \{(x^1, y^1), \dots, (x^N, y^N)\}$

Find values of parameters that maximize the likelihood of the observed training data: Maximum likelihood estimation (MLE)

$$\max_w \sum_{i=1}^N \log P(y_i | x^i, w)$$

$$= \max_w \sum_{i=1}^N \log e^{\frac{-(y_i - w^T x^i)^2}{2\sigma^2}}$$

$$= \max_w \sum_{i=1}^N \frac{-(y_i - w^T x^i)^2}{2\sigma^2} - N \log \sqrt{2\pi} \sigma$$

$\Rightarrow \min_w \sum_{i=1}^N (y_i - w^T x^i)^2$ $\left| \begin{array}{l} \sigma \text{ is fixed.} \\ \text{least square loss.} \end{array} \right.$

Controlling Overfitting: Bayesian estimation

- Parameters are not single points but random variables that follow a distribution called the prior distribution $p_0(w)$

$$w \sim \underline{N_d}(w; \underbrace{0_{d \times 1}}_{\mu}, \underbrace{\sigma_0^2 I_{d \times d}}_{\Sigma}) = \frac{1}{(2\pi)^{d/2} \sigma} e^{-\frac{w^T w}{2\sigma_0^2}}$$
$$= \frac{1}{(2\pi)^{d/2} \sigma} e^{-\frac{\|w\|^2}{2\sigma_0^2}}$$

- Training objective: Bayesian estimation

- Model defines distribution of label given parameters and x .

$D \sim i.i.d$ observations $\{(x^i, y^i) : i = 1 \dots N\}$

$$P(\underline{D} | \underline{w}) = \prod_{i=1}^N P(y_i | x^i, w)$$

During training, compute the posterior distribution of parameters given D & prior distribution.

$$P(w|D, p_0) = \frac{P(D|w) p_0(w)}{\int_{w'} P(D|w') p_0(w') dw'}$$

During Inference: given a test instance x we need to calculate.

$$P(y|x, D, p_0) = \int \underbrace{P(y|x, w)}_p \underbrace{P(w|D, p_0)}_p dw$$

A common approximation: $\nearrow$ W often intractable. $\searrow$ posterior.

During training find the mode of $P(w|D, p_0)$

$\hat{W} = \underset{w}{\operatorname{argmax}} P(w|D, p_0)$ & MAP = maximum estimate a-posteriori estimate

During inference

$$p(y|x, D) \approx p(y|x, \hat{w})$$

Prior probabilities for w for LR

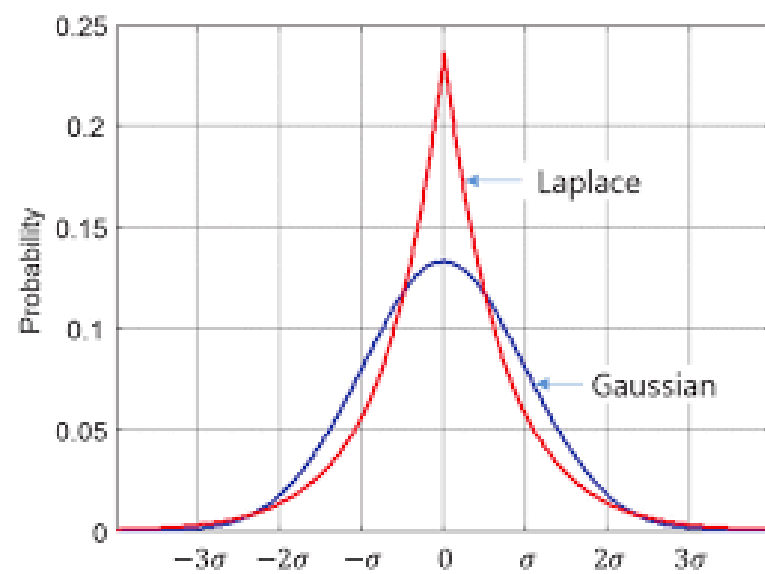
1) Gaussian
$$p_0(w) = \frac{e^{-\frac{\|w\|^2}{2\sigma^2}}}{(2\pi)^{d/2} \sigma}$$

It can be shown that MAP with this $p_0(w)$ gives rise to Ridge regression.

② Prior probability distribution is Laplace.

$$p_0(w) = \frac{1}{2b} e^{-\frac{\|w\|_1}{b}} \quad b > 0$$

→ It can be shown that the above MAP parameters are equivalent to Lasso objective.



Demo

- Colab notebook on linear regression:
<https://colab.research.google.com/github/jakevdp/PythonDataScienceHandbook/blob/master/notebooks/05.06-Linear-Regression.ipynb>
- 2-D regression plots:
[https://colab.research.google.com/github/rafiag/DI2020/blob/main/002a_Multi_Linear_Regression_\(EN\).ipynb](https://colab.research.google.com/github/rafiag/DI2020/blob/main/002a_Multi_Linear_Regression_(EN).ipynb)

Summary for linear regression

- Regression task: output a single real-value for any input.
- Choose a function class to serve as regression model.
- Represent input as a real-vector $x \in R^d$
- Parameters for linear regression model
 $w = [w_1, \dots, w_d]', w_0$ using which predicted value is defined as $f(x) = w \cdot x + w_0$
- To train: define an error function. Fit parameters to minimize error on the training data.